EE321

Computer Architecture

Course Presentation

Dr. Abdelhakim Khouas

Email : akhouas@hotmail.com

ab.khouas@univ-boumerdes.dz

IGEE (ex. INELEC)

University M'hamed Bougara of Boumerdes

Lecture Outline

1. Teacher
2. Course objectives
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5. Prerequisite
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7. Evaluation
8. Other books

1. Teacher

Pr. Abdelhakim Khouas

- Office: B428
- Email: akhouas@hotmail.com, ab.khouas@univ-boumerdes.dz
- Consultation: appointment by email
- Office hours: Sunday and Tuesday 11h30-13h00

2. Course objectives

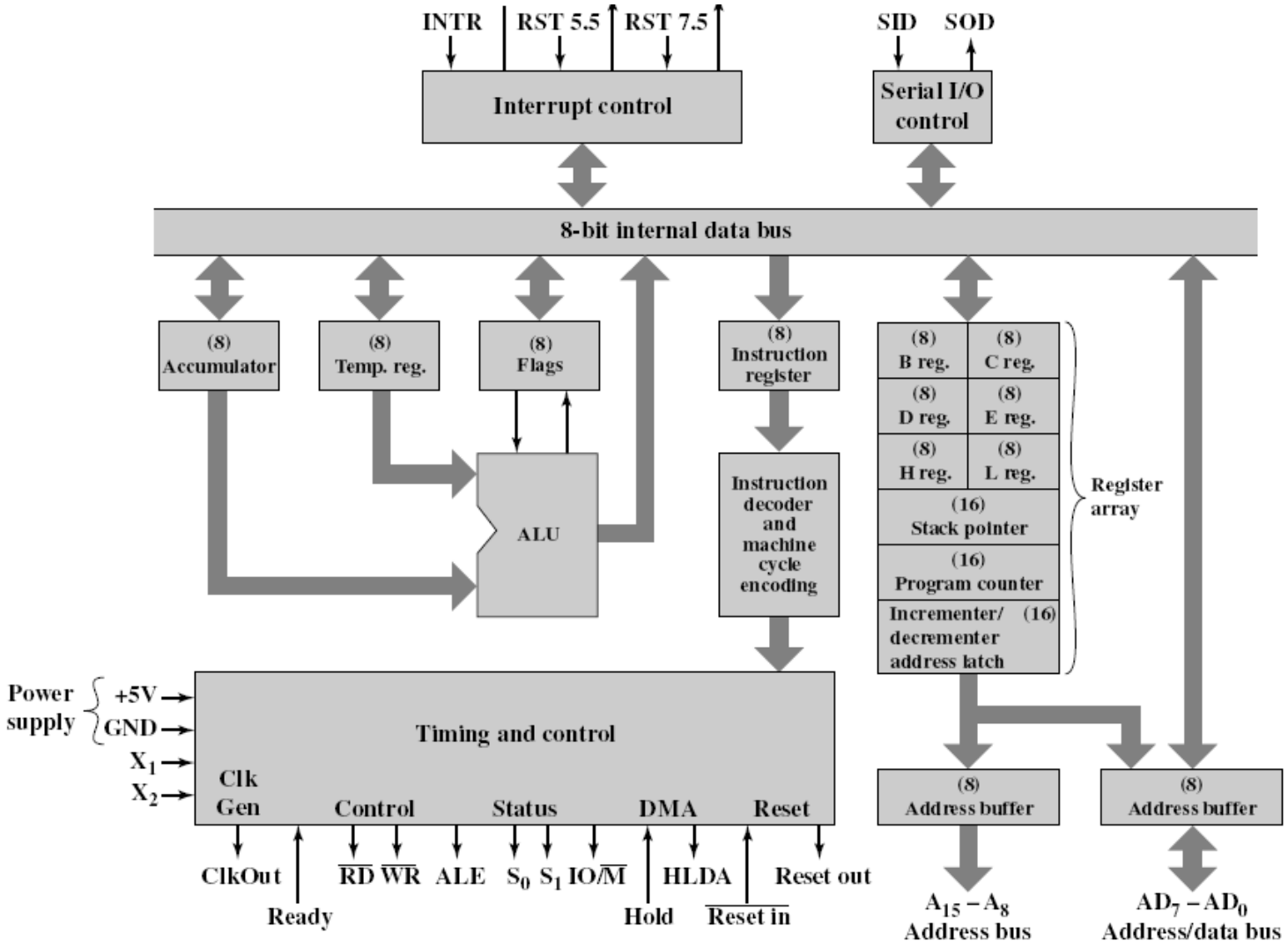
Learn structure and function of computers and understand how computer systems work

A computer consists of a set of component; computer architecture describes the way in which components are interconnected and function

Why learn computer architecture?

- Understand how codes (programs) are executed by the computer
- Write optimized programs (codes)
- Use efficiently microprocessors and microcontrollers

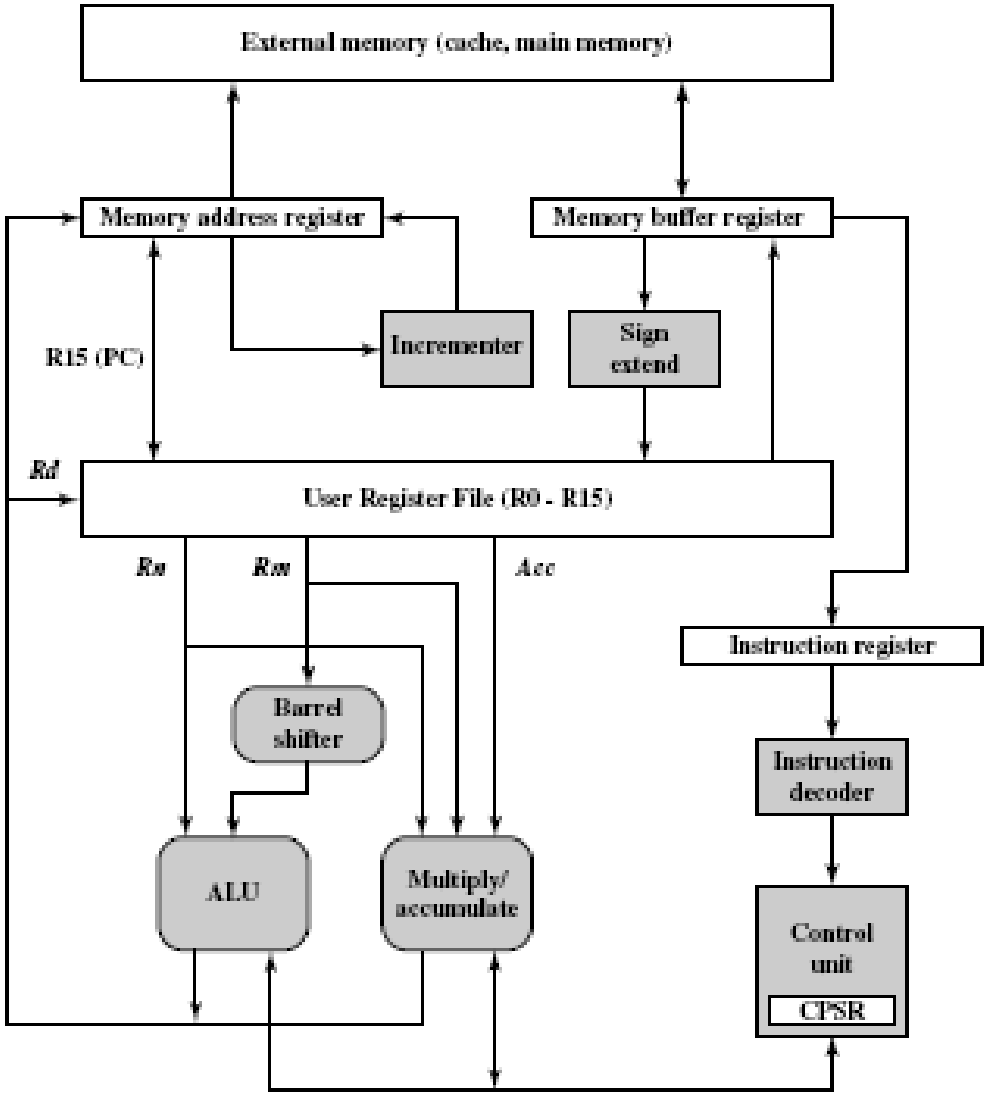
2. Course objectives



Intel 8085 CPU block diagram

Source: Computer Organization and Architecture, by W. Stallings

2. Course objectives



Simplified ARM processor organization

Source: Computer Organization and Architecture, by W. Stallings

3. Textbook

Computer Organization and Architecture,

By William Stallings

- Edition: 8th (last edition is 10th, 2016)
- Publisher: Prentice Hall
- Year: 2010

4. Course Outline

1. Review of Digital Design
2. Top Level of Computer
3. Central Processing Unit (CPU)
4. Control Unit
5. Memory
6. Instruction Set and Addressing Modes
7. Input/output Devices

5. Prerequisite

The only prerequisite I assume is: Digital Design

- Number representation (signed numbers)
- Boolean Logic
- Logic gates
- Combinational circuit (Multiplexer/Demultiplexer, Tristate buffer, Encoder/Decoder, Adder/ Subtractor, Arithmetic and Logic Unit (ALU)
- Latches, Flip-Flops (DFF), and Registers

6. Course Websites

Course Website

- http://akhouas.univ-boumerdes.dz/index.php/computer-architecture/

Google Classroom

- Class code : **o723viat**

Content

- Course description
- Chapter slides
- Previous control and final exams with solutions
- Documentation

7. Evaluation

Control
- 40 %

Final exam
- 60 %

Quizzes
- Bonus

Absences
- **Attendance is mandatory**
- More than **10** absences will result in **exclusion**

Knowledge and information is not power, knowing how to use knowledge and information is the power
- Only understanding questions, learning by heart is useless

8. Other books

[1] *"Computer Organization and Design: The Hardware/Software Interface"*, by D. A. Patterson and J. L. Hennessy

[2] *"Computer Architecture: A Quantitative Approach"*, by J. L. Hennessy and D. A. Patterson

[3] *"Digital Design and Computer Architecture"* by David Money Harris and Sarah L. Harris